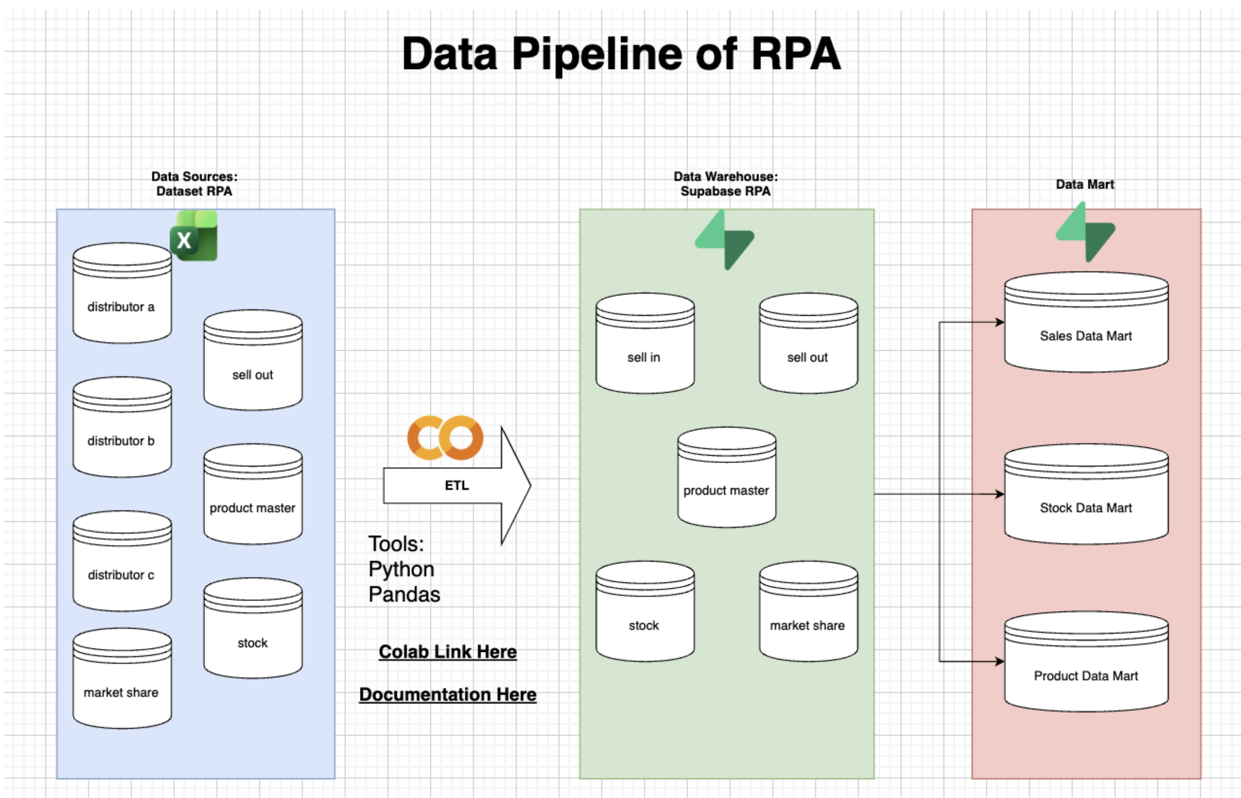
Data Pipeline Guide

This document explains how our data is prepared before it is loaded into the Supabase database. It covers the cleaning steps and how each final table is built from the raw sources.

1. Pipeline Summary

Detail	What it means
Code Source Notebook	https://colab.research.google.com/drive/10jxkRk4BPZScEI_QoSAFT6I5JJ102__w?usp=sharing
Raw Data File URL	The Google Sheets URL used by gdown.
Final Database	PostgreSQL tables in **Supabase**.



2. Source Data (Inputs)

All data is read from a single multi-sheet Excel file.

Data Source	What it Tracks	How Often (Granularity)
Sell-In (A, B, C)	Shipments from each distributor to stores.	Daily
Sell-Out	Products are sold from stores to customers.	Daily
Stock	Inventory counts (stock at start of day).	Daily Snapshot
Market Share	External data on brand performance.	Weekly Snapshot
Product Master	Details about each product (SKU).	Dimension

3. Cleaning & Prep Steps (Stage 1)

Before combining the data, we clean the raw files:

A. Quality Checks

- We check every raw sheet for **Missing Values (Nulls)** and **Duplicate Records**.

B. Product Master Fix (raw_product_master)

- The raw list had errors for certain products (SKU22 and SKU25).
- We check the right price from the transaction data in the sell_out table.
- Index 22 and 26 are the wrong price.
- **Action:** We permanently **delete** the invalid rows (index 22 and 26) to ensure the product master list is unique and correct.

C. Formatting

- The week_date column in the Market Share data is converted to a proper date format.
- The prefix "brand:" is removed from the brand column in the Market Share data.

4. The Final Tables (Outputs)

The pipeline creates four simple, clean tables in Supabase.

4.1 Table: product_master

Purpose: A simple lookup table for all product details.

Column	Data Type	Key Details
sku_id	Text (Primary Key)	The product code.
product_name	Text	The Shelf Product Name
brand	Text	The Brand of the product
format	Text	The type of product
size	Text	The size, usually in mL
price	Number (Int)	Price of product in Rp

How It's Made:

1. We start with the raw raw_product_master sheet.
2. For the duplicates, there are 2 SKUs with different pricing. To find the real price, we can check the sell_out table to ensure price is the same as stated in the transactions.
3. We drop the incorrect rows identified during cleaning.

	format	product_name	size	sku_id	brand	price
15	LIQUID SOAP	RELAXING HAIR AND BODY WASH 30 ML	30 ML	SKU16	Venus	7350
16	MOISTURIZER	ACNE MOISTURIZER 30ML	30 ML	SKU17	Venus	7350
17	MOISTURIZER	BRIGHTENING MOISTURIZER 30ML	30 ML	SKU18	Venus	7350
18	MOISTURIZER	ENERGIZING MOISTURIZER 30ML	30 ML	SKU19	Venus	7350
19	SERUM	ACNE SERUM 25ML	25 ML	SKU20	Venus	120490
20	SERUM	BRIGHTENING SERUM 25ML	25 ML	SKU21	Venus	120490
21	SERUM	ENERGIZING SERUM 25ML	25 ML	SKU22	Venus	120490
23	SUNSCREEN	ACNE SUNSCREEN SPF50 30 ML	30 ML	SKU23	Venus	39700
24	SUNSCREEN	BRIGHTENING SUNSCREEN SPF50 30 ML	30 ML	SKU24	Venus	39700
25	SUNSCREEN	TRIPLE PROTECTION SUNSCREEN SPF50 30 ML	30 ML	SKU25	Venus	39700

4.2 Table: total_sell_ins

Purpose: A single, complete list of all shipments from all distributors to the stores.

How It's Made:

1. **Tagging:** We add a distributor column (A, B, or C) to each of the three Sell-In files.
2. **Combining:** We stack all three distributor files into one big table.
3. **Enriching:** We add product details (format, product_name, brand, etc.) by matching the sku_id to the cleaned product_master table.

Column	Data Type	Key Details
date	Date	The day the product was shipped.
store_id	Text	The store that received the shipment.
sku_id	Text	The product code.
distributor	Text	Which distributor handled the shipment.
qty	Number (Int)	Quantity shipped.
value	Number (Int)	Total monetary value of the shipment.
format, product_name, size, brand	Text	Product details added from product_master.

	date	store_id	sku_id	distributor	format	product_name	size	brand	qty	value
0	2025-07-05	STORE1	SKU1	distributor_a	FACIAL WASH	ACNE FACE WASH 500 ML	500 ML	Venus	4	528360
1	2025-07-09	STORE1	SKU1	distributor_a	FACIAL WASH	ACNE FACE WASH 500 ML	500 ML	Venus	2	264180
2	2025-07-10	STORE1	SKU1	distributor_a	FACIAL WASH	ACNE FACE WASH 500 ML	500 ML	Venus	1	132090
3	2025-07-11	STORE1	SKU1	distributor_a	FACIAL WASH	ACNE FACE WASH 500 ML	500 ML	Venus	1	132090
4	2025-07-12	STORE1	SKU1	distributor_a	FACIAL WASH	ACNE FACE WASH 500 ML	500 ML	Venus	2	264180

4.3 Table: stock

Purpose: A daily record of how inventory changes at each store.

How It's Made:

1. **Calculate Stock In:** We total up all the shipments (Sell-In) for each product at each store for the day.
2. **Starting Point:** We use the raw_stock (Beginning Stock) sheet as the base.
3. **Merging In/Out:** We add columns for **Stock Out** (from Sell-Out data) and **Stock In** (from our calculated total Sell-In). If there was no activity, we use 0.
4. **Calculating End Stock:** We calculate the final end-of-day stock count using the formula: End Stock = Beginning Stock - Stock Out + Stock In
5. **Enriching:** We add product details from the product_master table.
6. **Casting:** All stock columns are converted to whole numbers (int).
7. **Standardizing:** Transformed all negative data into 0 because The data source treats negative values as 0.

Column	Data Type	Key Details
date, store_id, sku_id	Date, Text, Text	Defines the record.
beginning_stock	Number (Int)	Inventory at start of day.
stock_out	Number (Int)	Products sold to customers (from Sell-Out).
stock_in	Number (Int)	Products received from distributors (from Sell-In).
end_stock	Number (Int)	**Calculated** inventory at end of day.
format, product_name, size, brand	Text	Product details added from product_master.

date	store_id	sku_id	format	product_name	size	brand	beginning_stock	stock_out	stock_in	end_stock
2025-07-01	STORE1	SKU1	FACIAL WASH	ACNE FACE WASH 500 ML	500 ML	Venus	7	2	0	5
2025-07-02	STORE1	SKU1	FACIAL WASH	ACNE FACE WASH 500 ML	500 ML	Venus	5	0	0	5
2025-07-03	STORE1	SKU1	FACIAL WASH	ACNE FACE WASH 500 ML	500 ML	Venus	5	0	0	5
2025-07-04	STORE1	SKU1	FACIAL WASH	ACNE FACE WASH 500 ML	500 ML	Venus	5	3	0	2
2025-07-05	STORE1	SKU1	FACIAL WASH	ACNE FACE WASH 500 ML	500 ML	Venus	2	0	4	6

4.4 Table: market_share

Purpose: A clean list of weekly brand market share data.

How It's Made:

1. We start with the raw raw_market_share sheet.
2. We clean up the dates and remove the "brand:" prefix from the brand names.

Column	Data Type	Key Details
week_date	Date	The start date of the reporting week.
brand	Text	The brand name.
share	Decimal (Float)	The market share percentage.

	week_date	brand	format	share
0	2025-07-07	Venus	FACIAL WASH	0.290176
1	2025-07-07	Mars	FACIAL WASH	0.248471
2	2025-07-07	Jupiter	FACIAL WASH	0.231506
3	2025-07-07	Saturnus	FACIAL WASH	0.109679
4	2025-07-07	Uranus	FACIAL WASH	0.120167

4.5 Table: sell_out

Purpose: A transaction-level record of products sold to customers, enriched with calculated unit price and product master details.

How It's Made:

1. **Rename Unit Price:** The original value column in the raw_sell_out sheet is renamed to price (representing the unit sale price).
2. **Calculate Total Value:** A new value column is calculated by multiplying the qty by the new price column.
3. **Enriching:** The modified raw_sell_out table is Left-Joined with the cleaned product_master table on sku_id.
4. **Final Schema:** Columns are reordered to match the final structure, and the temporary price column is renamed back to price.

Column	Data Type	Key Details
date	Date	The day the product was sold.
store_id	Text	The store where the sale occurred.
sku_id	Text	The product code.
format, product_name, size, brand	Text	Product details added from product_master.
qty	Number (Int)	Quantity sold in the transaction.
price	Number (Int)	Unit Sale Price (From original value column).
value	Number (Int)	Total Transaction Value (Calculated as qty * price).

5. Data Mart

5.1 Table: looker_sales_data_mart (Reporting Mart)

Purpose: A specialized mart built directly from the sell_out transactional data, denormalized with comprehensive date and time dimensions for flexible reporting and visualization in BI tools.

How It's Made:

1. **Base Table:** Starts with the enriched sell_out table (4.5).
2. **Date Dimensions:** The following derived columns are added for reporting simplicity:
    - week_date: The start date of the ISO week.
    - iso_week: The ISO week number of the year.
    - month: The month number (1-12).
    - month_name: The full name of the month.
    - day: The day of the month.
    - day_name: The full name of the day.

Column	Data Type	Key Details
date	Date	The original sale date.
store_id	Text	The store where the sale occurred.
sku_id	Text	The product code.
format, product_name, size, brand	Text	All product details are included.
qty	Number (Int)	Quantity sold in the transaction.
price	Number (Int)	Unit Sale Price.
value	Number (Int)	Total Transaction Value.
week_date	Date	Start date of the week.
iso_week	Number (Int)	ISO Week Number.
month	Number (Int)	Month Number.
month_name	Text	Month Name.
day	Number (Int)	Day of the month.
day_name	Text	Day of the week name.

date date	store_id text	sku_id text	format text	product_name text
2025-07-01	STORE6	SKU10	FACIAL WASH	ENERGIZING FACE WASH
2025-07-01	STORE4	SKU22	SERUM	ENERGIZING SERUM 25ML
2025-07-01	STORE4	SKU15	LIQUID SOAP	RELAXING HAIR AND BOD
2025-07-01	STORE6	SKU9	FACIAL WASH	ENERGIZING FACE WASH
2025-07-01	STORE2	SKU11	FACIAL WASH	ENERGIZING FACE WASH

5.2 Table: looker_stock_data_mart (Reporting Mart)

Purpose: A specialized mart for inventory (stock) reporting, pre-calculated with time dimensions and the monetary value of products moved out of stock (sold) for simplified analysis of inventory costs and potential revenue.

How It's Made:

1. **Base Table:** Starts with the stock table (4.3).
2. **Product Enrichment: Left-Joins** with the product_master table (4.1) on sku_id to include the unit price.
3. **Date Dimensions:** Adds date dimensions derived from the date column: iso_week, month, and month_name.
4. **Calculated Metric:** Computes value_stock_out_weekly by multiplying stock_out (quantity sold) by the product_master's price.

Column	Data Type	Key Details
date	Date	The day of the stock record.
store_id	Text	The store identifier.
sku_id	Text	The product code.
beginning_stock	Number (Int)	Inventory at start of day.
stock_out	Number (Int)	Products sold to customers (from

		Sell-Out).
stock_in	Number (Int)	Products received from distributors (from Sell-In).
end_stock	Number (Int)	Calculated inventory at end of day.
iso_week	Number (Int)	ISO Week Number.
month	Number (Int)	Month Number.
month_name	Text	Month Name.
value_stock_out_weekly	Number (Float)	Monetary value of the stock that was sold.

5.3 Table: looker_market_share_data_mart

Purpose: A specialized mart for market share reporting, pre-calculated with time dimensions for flexible weekly analysis and comparison in BI tools.

How It's Made:

1. **Base Table:** Starts with the market_share table (4.4).
2. **Date Dimensions:** The following derived columns are added for reporting simplicity:
   - iso_week: The ISO week number of the year.
   - month: The month number (1-12).
   - month_name: The full name of the month.

Column	Data Type	Key Details
week_date	Date	The start date of the reporting week.
brand	Text	The brand name.
share	Decimal (Float)	The market share percentage.
iso_week	Number (Int)	ISO Week Number.
month	Number (Int)	Month Number.
month_name	Text	Month Name.